

| Question |     |      |  | Marking details                                                                                                                                                                                  | Marks available |          |          |          |          |          |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                  | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 6        | (a) | (i)  |  | This increases the voltage <b>or</b> decreases the current (1)<br>reducing energy loss as heat [in the wires] <b>or</b> more efficient (1)<br>Don't accept prevents heat loss                    | 2               |          |          | 2        |          |          |
|          |     | (ii) |  | $1800 \times 10^6 = I \times 400 \times 10^3$ (1)<br>$I = \frac{1800 \times 10^6}{400 \times 10^3}$ (1)<br>$I = 4500$ [A] (1)<br>Award 2 marks for answers of $4.5 \times 10^n$ where $n \neq 3$ | 1               | 1<br>1   |          | 3        | 3        |          |
|          | (b) |      |  | Between 0:00 and 6:00 <b>or</b> 1:00 and 5:00 <b>or</b> 1:00 and 6:00 <b>or</b> 0:00 and 5:00 (1)<br>when demand is <u>lowest</u> (1)                                                            |                 | 2        |          | 2        |          |          |
|          |     |      |  | <b>Question 6 total</b>                                                                                                                                                                          | <b>3</b>        | <b>4</b> | <b>0</b> | <b>7</b> | <b>3</b> | <b>0</b> |